

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1104

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)
Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:35

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I __S.Poojitha(192320076)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Poojitha

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving

Q1. Scenario: Online Banking System

System Components:

Account Management, Fund Transfer, Bill Payment, Transaction History

Given Constraints:

- Transaction completion time < 2 seconds
- Multi-factor authentication required
- Support 5000+ concurrent users

Question:

Analyze the system using quality assurance techniques under the given constraints. Identify potential defects related to security, performance, and reliability. Propose suitable testing strategies and justify how they ensure system quality. **(BL4)**

Ans) Q1. Online Banking System – Quality Assurance

An online banking system must be **secure, fast, reliable, and scalable**. The main quality requirements are **transaction time below 2 seconds, mandatory MFA, and support for 5000+ concurrent users**.

1. Potential Defects

- **Security:** Weak passwords, MFA bypass, SQL injection, session hijacking, and unauthorized access.
- **Performance:** Transaction response time exceeding 2 seconds, slow database queries, and system slowdown with 5000+ users.
- **Reliability:** Duplicate fund transfers, incorrect balance updates, failed transactions, or money deducted without being credited.

2. Testing Strategies

Testing	Purpose
Functional Testing	Verify account management, fund transfer, bill payment, and transaction history.
Security Testing	Test MFA, authentication, authorization, SQL injection, and session security.
Load Testing	Check performance with 5000+ concurrent users.
Stress Testing	Test system beyond 5000 users to find its breaking point.
Performance Testing	Ensure transactions complete in less than 2 seconds .

Reliability Testing	Test server/network failures and ensure transactions are recovered correctly.
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3. Justification

These techniques ensure that the banking system:

- Protects sensitive customer information.
- Prevents unauthorized transactions.
- Handles **5000+ users** efficiently.
- Completes transactions within **2 seconds**.
- Maintains accurate balances and transaction records.
- Recovers safely from failures.

Conclusion: A combination of **functional, security, performance, load, stress, and reliability testing** ensures that the online banking system meets its quality requirements.

Q2. Scenario: E-Commerce Shopping Application

System Features:

Product Search, Shopping Cart, Checkout, Payment Gateway

Usability Constraints:

- Checkout process within 3 steps
- Easy product navigation
- Responsive design for mobile and desktop devices

Question:

Apply usability testing techniques to evaluate the application. Identify usability challenges and recommend design improvements that satisfy the given constraints. Explain how the proposed changes improve customer satisfaction and efficiency. **(BL4)**

ANS) Usability Testing Techniques:

- **User Testing:** Observe users while they search, add products to the cart, and complete payment.
- **Task Analysis:** Check how many steps and clicks are required to complete shopping.
- **A/B Testing:** Compare different layouts, buttons, and checkout designs.
- **Responsive Testing:** Test the application on mobile, tablet, and desktop screens.

Usability Challenges:

- Difficult product navigation or unclear categories.
- Too many steps during checkout.
- Complicated payment forms.
- Poor mobile screen layout.
- Important buttons such as **“Add to Cart”** or **“Buy Now”** may be difficult to find.

Design Improvements:

1. Use clear categories, search bar, filters, and sorting options for easy navigation.
2. Design checkout in **3 steps or fewer**:
 - Step 1: Address
 - Step 2: Payment
 - Step 3: Order Confirmation
3. Use large, clearly visible **Add to Cart** and **Checkout** buttons.
4. Use responsive layouts that automatically adjust to mobile and desktop screens.
5. Provide progress indicators and clear error messages during checkout.

Benefits:

- Fewer checkout steps reduce **time and effort**.
 - Simple navigation helps customers find products quickly.
 - Responsive design improves the experience on all devices.
 - Clear feedback and error messages reduce confusion.
 - Overall, these improvements increase **customer satisfaction, usability, and purchase completion rates**.
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Q3. Scenario: Hospital Management System

Modules:

Patient Registration, Appointment Scheduling, Medical Records, Billing

QA Constraints:

- Patient data accuracy must be maintained
- System availability 24/7
- Simultaneous access by multiple departments

Question:

Use quality assurance practices to analyze the system. Identify critical test cases and discuss how the constraints influence testing priorities, defect prevention, and overall system reliability. **(BL5)**

QA Practices:

- **Functional Testing:** Verify patient registration, appointments, medical records, and billing.
- **Data Validation Testing:** Ensure patient details, medical history, and billing data are accurate.
- **Performance Testing:** Test the system with multiple departments accessing it simultaneously.
- **Security Testing:** Protect sensitive patient information from unauthorized access.
- **Availability Testing:** Ensure the system works continuously, 24/7.

Critical Test Cases:

1. Register a patient with valid and invalid details.
2. Prevent duplicate patient records.

3. Book, reschedule, and cancel appointments correctly.
4. Verify that medical records are saved and retrieved without data loss.
5. Check that billing calculations and payments are accurate.
6. Test simultaneous access by doctors, billing staff, and receptionists.
7. Test system recovery after server/network failure.
8. Verify that only authorized users can access patient records.

Impact of Constraints:

- **Data accuracy:** Requires strong validation, database testing, and duplicate-data checks.
- **24/7 availability:** Requires backup, recovery, failover, and continuous monitoring tests.
- **Multiple departments:** Requires concurrency and load testing to prevent conflicts and slowdowns.

Overall Reliability:

These QA practices help prevent critical defects, protect patient data, reduce system downtime, and ensure that hospital staff can access **accurate information quickly and reliably**.

Q4. Scenario: Smart Library Management System

Features:

Book Search, Book Reservation, Borrow/Return, Notifications

Usability Constraints:

- Search results displayed within 2 seconds
- Accessible interface for all age groups
- Minimal learning curve for first-time users

Question:

Conduct a usability evaluation using appropriate testing methods. Analyze how the given constraints impact interface design and recommend improvements to enhance usability and accessibility. **(BL5)**

Ans) Usability Testing Methods:

- **User Testing:** Observe first-time users while searching, reserving, and borrowing books.
- **Task Analysis:** Measure the number of steps needed to complete common tasks.
- **Performance Testing:** Check whether search results appear within **2 seconds**.
- **Accessibility Testing:** Test the interface with different age groups and users with accessibility needs.

Usability Challenges:

- Complicated menus may confuse first-time users.
- Small text or buttons can be difficult for elderly users.
- Slow search results can reduce user satisfaction.
- Unclear reservation and notification messages may cause confusion.

Design Improvements:

1. Provide a **simple search bar** with filters such as author, title, and category.

2. Optimize the database and search system to display results within **2 seconds**.
3. Use **large fonts, clear icons, high contrast, and readable labels**.
4. Keep navigation simple with options like **Search, Reserve, Borrow/Return, and Notifications**.
5. Provide clear confirmation messages after reservation or return.
6. Use tooltips or short instructions to help first-time users.

Benefits:

- Faster search improves **efficiency**.
 - Simple navigation reduces the **learning curve**.
 - Accessible fonts, colors, and controls support **all age groups**.
 - Clear feedback reduces errors and confusion.
 - Overall, the improvements provide a **simple, fast, and accessible library experience**.
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Q5. Scenario: Airline Reservation System

Modules:

User Login, Flight Search, Seat Selection, Ticket Booking

QA & Usability Constraints:

- No booking duplication
- System must support peak holiday traffic
- Booking interface should be simple and error-free

Question:

Apply both quality assurance and usability testing techniques under the specified constraints. Identify major risks, propose suitable testing methods, and explain how the system can be optimized for reliability, performance, and user experience. **(BL4)**

Ans)Major Risks:

- Duplicate booking due to simultaneous users.
- System slowdown or failure during peak holiday traffic.
- Incorrect seat availability or double seat selection.
- Login and payment errors.
- Complicated booking interface causing user mistakes.

Testing Methods:

- **Functional Testing:** Test login, flight search, seat selection, and ticket booking.
- **Concurrency Testing:** Check multiple users booking the same flight/seat simultaneously.
- **Load & Stress Testing:** Test the system with high traffic during holidays.
- **Security Testing:** Verify secure login and protection of customer information.
- **Usability Testing:** Observe users completing bookings and identify confusing screens or steps.
- **Recovery Testing:** Check whether bookings remain correct after server or network failures.

Design & System Improvements:

1. Use **real-time seat locking** to prevent two users from selecting the same seat.
2. Use database **transactions and unique booking IDs** to prevent duplicate bookings.
3. Use **load balancing and scalable servers** to handle holiday traffic.
4. Keep the booking process simple with clear steps, labels, and error messages.
5. Show real-time seat availability and provide confirmation before final booking.
6. Optimize database queries and caching to improve response time.

Benefits:

- Prevents duplicate and incorrect bookings.
- Handles high traffic with better **performance and reliability**.
- Simple navigation reduces user errors.
- Faster and clearer booking improves **customer satisfaction and efficiency**.